

Computing limits of functions



Outline

Algebra of limits theorem

Sign-preservation theorems

Comparison theorem

Partial arithmetization of the infinity symbol

Change of variable theorem for limits

Algebra of limits theorem



Introduction: Algebra of limits theorem

- ▶ We state the theorems on limits of functions that follow immediately from the corresponding theorems on limits of sequences and from the **sequential definition of limit**
- ▶ The first tool is the **algebra of limits theorem**: the limit of a sum, product and quotient is computed from the limits of the single terms or factors
- ▶ The proof strategy will always be the same: we take any sequence $x_n \rightarrow c$, $x_n \neq c$, and apply the analogous theorem for sequences

Theorem 1: Algebra of limits, case of finite limits

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell_1 \in \mathbb{R}, \quad 2. g(x) \rightarrow \ell_2 \in \mathbb{R}.$$

Thesis as $x \rightarrow c$:

$$1. f(x) \pm g(x) \rightarrow \ell_1 \pm \ell_2, \quad 2. f(x) g(x) \rightarrow \ell_1 \ell_2,$$

$$3. \frac{f(x)}{g(x)} \rightarrow \frac{\ell_1}{\ell_2}$$

$$(\ell_2 \neq 0, g(x) \neq 0, \text{ eventually as } x \rightarrow c).$$

Proof

Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \quad \text{and} \quad x_n \rightarrow c \quad \text{as} \quad n \rightarrow +\infty.$$

By hypothesis we have

$$f(x_n) \rightarrow \ell_1 \quad \text{and} \quad g(x_n) \rightarrow \ell_2 \quad (\ell_1, \ell_2 \in \mathbb{R}).$$

From the theorem on the algebra of limits for sequences we therefore conclude that

$$f(x_n) \pm g(x_n) \rightarrow \ell_1 \pm \ell_2,$$

and hence

$$f(x) \pm g(x) \rightarrow \ell_1 \pm \ell_2.$$

Claims 2 and 3 of the theorem are proved in a perfectly analogous way. □

Summary

Algebra of limits:

- ▶ If $f(x) \rightarrow \ell_1 \in \mathbb{R}$ and $g(x) \rightarrow \ell_2 \in \mathbb{R}$ as $x \rightarrow c$, then sum, difference, product and quotient (with $\ell_2 \neq 0$) tend to the corresponding values
- ▶ The proof uses the sequential definition of limit and the analogous theorem for sequences

Sign-preservation theorems

Introduction: Sign-preservation theorems

- ▶ In this section we see that the **sign of the limit** is transferred to the function, at least eventually, and vice versa
- ▶ We will see two forms of the theorem, which follow from the corresponding theorems for sequences
- ▶ For continuous functions we will also obtain a specific version: if $f(c) > 0$, then f stays positive near c



Convention

In the following statements ℓ and c will be points of $\mathbb{R}^*$ unless otherwise stated.

Sign preservation (1^{st} form)

Theorem 2: Sign preservation, 1^{st} form

Hypotheses:

$$1. f(x) \rightarrow \ell \text{ as } x \rightarrow c, \quad 2. \ell \leq 0.$$

Thesis:

$$f(x) \leq 0 \text{ eventually as } x \rightarrow c.$$

Proof

Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \quad \text{and} \quad x_n \rightarrow c \quad \text{as} \quad n \rightarrow +\infty.$$

By the hypothesis we have

$$f(x_n) \rightarrow \ell > 0$$

hence, by the sign-preservation theorem for sequences, applied to the sequence $\{f(x_n)\}$, we conclude that

$$f(x_n) > 0, \quad \text{eventually.}$$

Since this holds for every sequence such that $x_n \rightarrow c$, we conclude that

$$f(x) > 0, \quad \text{eventually, as} \quad x \rightarrow c.$$



Sign preservation (2^{nd} form)

Theorem 3: Sign preservation for functions, 2^{nd} form

Hypotheses:

1. $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow c$,
2. $f(x) \geq 0$ eventually as $x \rightarrow c$.

Thesis:

$$\ell \geq 0.$$

Proof

It follows from the corresponding theorem for sequences.



Sign preservation for continuous functions

- For functions we also have the following sign-preservation theorem.

Theorem 4: Sign preservation for continuous functions

Hypotheses:

1. f is continuous at $c \in \mathbb{R}$,
2. $f(c) > 0$.

Thesis:

$$f(x) > 0 \text{ eventually as } x \rightarrow c.$$

Proof

If f is continuous at c , then

$$f(c) = \lim_{x \rightarrow c} f(x)$$

hence the hypothesis $f(c) > 0$ means that

$$f(c) = \lim_{x \rightarrow c} f(x) > 0$$

and this, by the sign-preservation theorem (1st form), implies that

$$f(x) > 0 \text{ eventually as } x \rightarrow c.$$



Summary

Sign preservation:

- ▶ **1st form:** if $f(x) \rightarrow \ell \leq 0$, then $f(x) \leq 0$ eventually as $x \rightarrow c$
- ▶ **2nd form:** if $f(x) \geq 0$ eventually and $f(x) \rightarrow \ell \in \mathbb{R}$, then $\ell \geq 0$
- ▶ **Continuous functions:** if f is continuous at c and $f(c) > 0$, then $f(x) > 0$ eventually as $x \rightarrow c$

Comparison theorem

Introduction: Comparison theorem

- ▶ The **comparison theorem** allows us to compute the limit of a function “trapped” between two functions having the same limit
- ▶ We will derive two very useful **corollaries**: functions dominated in absolute value by infinitesimal functions, and the product of an infinitesimal function and a bounded function
- ▶ These tools apply even when the algebra of limits is not enough, for instance with $\sin \frac{1}{x}$ as $x \rightarrow 0$

Comparison theorem

Theorem 5: Comparison theorem

Hypotheses:

1. $f(x) \rightarrow \ell$ and $g(x) \rightarrow \ell$ as $x \rightarrow c$,
2. $f(x) \leq h(x) \leq g(x)$ eventually as $x \rightarrow c$.

Thesis:

$$h(x) \rightarrow \ell \text{ as } x \rightarrow c.$$

Proof (1/2)

Let $\{x_n\}$ be any sequence such that

$$x_n \neq c, \forall n, \quad \text{and} \quad x_n \rightarrow c \quad \text{as} \quad n \rightarrow +\infty.$$

We want to prove that

$$h(x_n) \rightarrow \ell \quad \text{as} \quad n \rightarrow +\infty.$$

Proof (2/2)

By hypothesis we know that:

$$\begin{aligned} f(x_n) &\leq h(x_n) \leq g(x_n), \text{ eventually,} \\ f(x_n) &\rightarrow \ell \text{ and } g(x_n) \rightarrow \ell, \text{ as } x_n \rightarrow c. \end{aligned}$$

Hence, by the comparison theorem for sequences applied to:

$$\{f(x_n)\}, \{h(x_n)\} \text{ and } \{g(x_n)\}$$

we conclude that

$$h(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty.$$



Corollary of the comparison theorem (part I)

Corollary 1: Of the comparison theorem (part I)

Hypotheses:

1. $g(x) \rightarrow 0$ as $x \rightarrow c$,
2. $|h(x)| \leq g(x)$ eventually as $x \rightarrow c$.

Thesis:

$$h(x) \rightarrow 0 \text{ as } x \rightarrow c.$$

Proof

It follows from the corresponding corollary for sequences.



Example (1/2)

Let us prove that:

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

We have

$$x \rightarrow 0 \text{ as } x \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ does not exist}$$

hence we cannot apply Theorem 1 on the algebra of limits for functions.

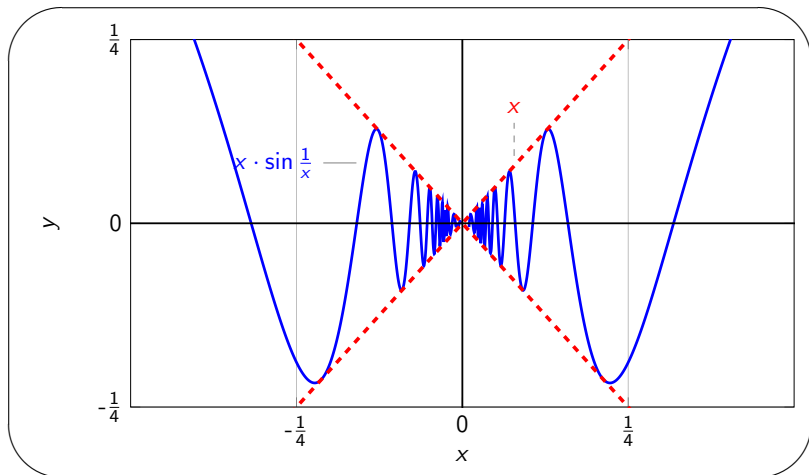
We have

$$\left| \sin \frac{1}{x} \right| \leq 1 \text{ hence } \left| x \sin \frac{1}{x} \right| \leq |x| \text{ and } |x| \rightarrow 0 \text{ as } x \rightarrow 0.$$

Hence, by Corollary 1 of the comparison theorem for functions, we have:

$$x \sin \frac{1}{x} \rightarrow 0 \text{ as } x \rightarrow 0.$$

Example (2/2)



Corollary of the comparison theorem (part II)

Corollary 2: Of the comparison theorem (part II)

Hypotheses:

1. $f(x) \rightarrow 0$ as $x \rightarrow c$,
2. $g(x)$ is bounded, eventually, as $x \rightarrow c$.

Thesis:

$$f(x)g(x) \rightarrow 0 \text{ as } x \rightarrow c.$$

Proof

It follows from the corresponding corollary for sequences.



Example (1/3)

Let us prove that:

$$\lim_{x \rightarrow +\infty} \frac{x + \sin x}{2x + \cos x} = \frac{1}{2}$$

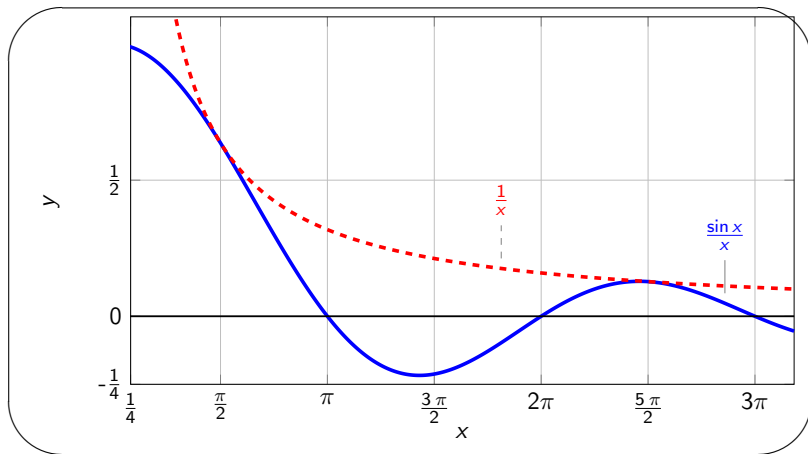
Factoring, we obtain

$$\frac{x + \sin x}{2x + \cos x} = \frac{x \left(1 + \frac{\sin x}{x}\right)}{2x \left(1 + \frac{\cos x}{2x}\right)} = \frac{1}{2} \left(\frac{1 + \frac{\sin x}{x}}{1 + \frac{\cos x}{2x}} \right)$$

By Corollary 2 of the comparison theorem for functions, we have:

$$\frac{\sin x}{x} \rightarrow 0 \quad \text{and} \quad \frac{\cos x}{2x} \rightarrow 0 \quad \text{as } x \rightarrow +\infty.$$

Example (2/3)



Example (3/3)

Then, by Theorem 1 on the algebra of limits, we therefore have:

$$\frac{x + \sin x}{2x + \cos x} = \frac{1}{2} \left(\frac{1 + \frac{\sin x}{x}}{1 + \frac{\cos x}{2x}} \right) \rightarrow \frac{1}{2} \cdot \frac{1 + 0}{1 + 0} = \frac{1}{2} \quad \text{as } x \rightarrow +\infty.$$

Comparison theorem and corollaries:

- ▶ If $f(x) \leq h(x) \leq g(x)$ eventually and $f(x), g(x) \rightarrow \ell$, then $h(x) \rightarrow \ell$
- ▶ If $|h(x)| \leq g(x)$ eventually and $g(x) \rightarrow 0$, then $h(x) \rightarrow 0$
- ▶ Infinitesimal times bounded is infinitesimal: for instance $x \sin \frac{1}{x} \rightarrow 0$ as $x \rightarrow 0$ and $\frac{\sin x}{x} \rightarrow 0$ as $x \rightarrow +\infty$

Partial arithmetization of the infinity symbol



Introduction: Partial arithmetization of ∞

- ▶ We extend the algebra of limits to the case where some limits are **infinite** or **zero**
- ▶ We will see for which operations the result is determined (for instance $\ell + \infty = \infty$, $\frac{\ell}{\infty} = 0$)
- ▶ As for sequences, the sign of ∞ is determined with the **rule of signs**

Theorem 6: Partial arithmetization of the infinity symbol (addition)

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell \in \mathbb{R}, \quad 2. g(x) \rightarrow +\infty, \quad 3. h(x) \rightarrow +\infty.$$

Thesis as $x \rightarrow c$:

1. $f(x) + g(x) \rightarrow \ell + \infty = +\infty,$
2. $f(x) - g(x) \rightarrow \ell - \infty = -\infty,$
3. $g(x) + h(x) \rightarrow +\infty + \infty = +\infty,$
4. $-g(x) - h(x) \rightarrow -\infty - \infty = -\infty.$

Proof

It follows from the corresponding theorem for sequences.



Theorem 7: Partial arithmetization of the infinity symbol (product)

Hypotheses as $x \rightarrow c$:

$$1. f(x) \rightarrow \ell \in \mathbb{R}, \quad 2. g(x) \rightarrow 0, \quad 3. h(x) \rightarrow \infty.$$

Thesis as $x \rightarrow c$:

$$\begin{aligned} 1. f(x) h(x) &\rightarrow \ell \cdot \infty = \infty \quad (\ell \neq 0), \\ 2. \frac{f(x)}{g(x)} &\rightarrow \frac{\ell}{0} = \infty \quad (\ell \neq 0), \quad 3. \frac{f(x)}{h(x)} \rightarrow \frac{\ell}{\infty} = 0. \end{aligned}$$

Proof

It follows from the corresponding theorem for sequences.



Example

- ▶ As for sequences, the **sign** of ∞ must be determined with the **rule of signs**.

$$\lim_{x \rightarrow -\infty} \left(\frac{1}{x} - 2 \right) x^3 = +\infty$$

Since

$$\left(\frac{1}{x} - 2 \right) \rightarrow -2 \quad \text{and} \quad x^3 \rightarrow -\infty \quad \text{as} \quad x \rightarrow -\infty.$$

Summary

Partial arithmetization of ∞ :

- ▶ $\ell \pm \infty = \pm\infty$, $+\infty + \infty = +\infty$, $-\infty - \infty = -\infty$
- ▶ $\ell \cdot \infty = \infty$ and $\frac{\ell}{0} = \infty$ if $\ell \neq 0$, $\frac{\ell}{\infty} = 0$
- ▶ The sign of ∞ is determined with the rule of signs

Change of variable theorem for limits



Introduction: Change of variable theorem for limits

- ▶ Many functions are **compositions** $f(g(x))$ of simpler functions
- ▶ The **change of variable theorem** allows us to compute the limit by setting $t = g(x)$: first we compute the limit t_0 of g , then the limit of $f(t)$ as $t \rightarrow t_0$
- ▶ It is one of the most used tools in computing limits: we will apply it to logarithms and exponentials

Change of variable in the limit

Theorem 8: Change of variable in the limit

Let f and g be two functions for which the composition $f \circ g$ is defined, at least eventually as $x \rightarrow x_0 \in \mathbb{R}^$. If:*

1. $\lim_{x \rightarrow x_0} g(x) = t_0 \in \mathbb{R}^*$, 2. $\lim_{t \rightarrow t_0} f(t) = \ell \in \mathbb{R}^*$
3. $g(x) \neq t_0$, eventually, as $x \rightarrow x_0$

Then:

$$\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t).$$

On hypothesis 3

Hypothesis 3. is not necessary when f is continuous at t_0 , or when $t_0 = \pm\infty$.

Proof (1/2)

Let $\{x_n\}$ be any sequence such that

$$x_n \neq x_0, \forall n, \quad \text{and} \quad x_n \rightarrow x_0 \quad \text{as} \quad n \rightarrow +\infty.$$

We have

$$g(x_n) \rightarrow t_0 \quad \text{as} \quad n \rightarrow +\infty \quad (\text{by hypothesis 1})$$

and

$$g(x_n) \neq t_0, \quad \text{eventually} \quad (\text{by hypothesis 3}).$$

Therefore

$$f(g(x_n)) \rightarrow \ell \quad (\text{by hypothesis 2}).$$

Proof (2/2)

If $t_0 = \pm\infty$ the condition $g(x) \neq \pm\infty$ is obviously satisfied, while if f is continuous at t_0 , $\ell = f(t_0)$, so if $g(x_n) = t_0$ for some n we would have

$$f(g(x_n)) = f(t_0) = \ell$$

and hence the convergence

$$f(g(x_n)) \rightarrow \ell \quad \text{as } n \rightarrow +\infty$$

would be guaranteed anyway.



Example (1/4)

Let us compute the limit

$$\lim_{x \rightarrow +\infty} \log \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right)$$

The functions f and g are:

$$g(x) = \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right) \quad \text{and} \quad f(t) = \log t.$$

We have $t = g(x)$ and

$$(f \circ g)(x) = f(g(x)) = \log \left(\frac{2x^3 + 4x + 1}{5(x+1)^3} \right) \quad \text{and} \quad x_0 = +\infty.$$

Example (2/4)

We compute

$$\lim_{x \rightarrow +\infty} g(x) = \frac{2}{5} \quad \text{and consequently} \quad t_0 = \frac{2}{5}.$$

Now we compute

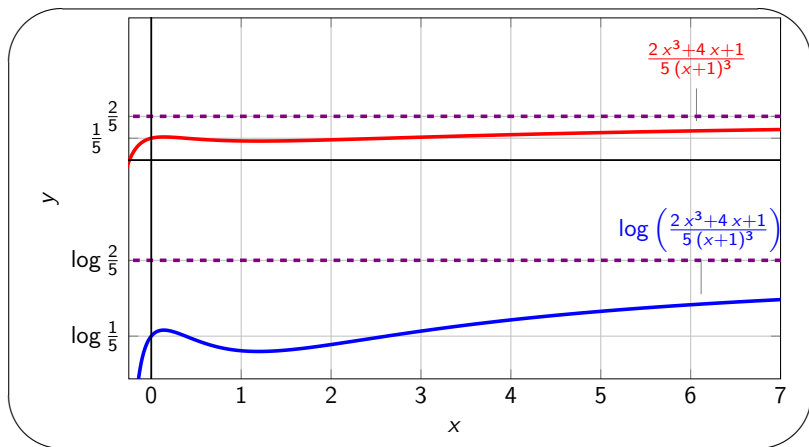
$$\lim_{t \rightarrow 2/5} \log t = \log \frac{2}{5}$$

(as we will see, the logarithm is continuous in $\mathbb{R}_+$)

Hence

$$\lim_{x \rightarrow +\infty} f(g(x)) = \log \frac{2}{5}$$

Example (3/4)



Example (4/4)

Moreover:

$$\lim_{x \rightarrow 0} f(g(x)) = \log \frac{1}{5}$$

using the theorem we have:

$$\lim_{x \rightarrow 0} g(x) = \frac{1}{5} \quad \text{and} \quad \lim_{t \rightarrow 1/5} \log t = \log \frac{1}{5}$$

since

$$\lim_{x \rightarrow 0} 2x^3 + 4x + 1 = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} 5(x+1)^3 = 5$$

Example (1/3)

Let us compute the limits

$$\lim_{x \rightarrow 0^+} e^{\frac{1}{x}} \quad \text{and} \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}}.$$

The functions f and g are:

$$g(x) = \frac{1}{x} \quad \text{and} \quad f(t) = e^t.$$

We have $t = g(x)$ and

$$(f \circ g)(x) = f(g(x)) = e^{\frac{1}{x}} \quad \text{moreover} \quad x_0 = 0^+ \quad \text{and} \quad x_0 = 0^-.$$

Example (2/3)

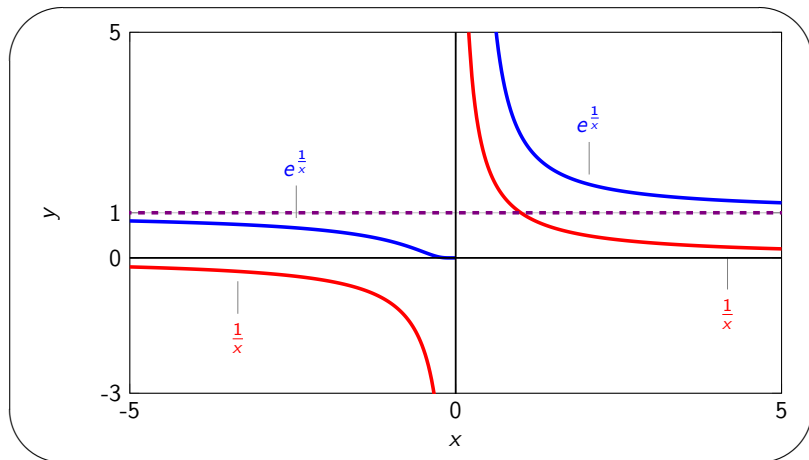
1. $\frac{1}{x} \rightarrow +\infty$ as $x \rightarrow 0^+$ and hence $t_0 = +\infty$. Now we compute:

$$\lim_{t \rightarrow +\infty} e^t = +\infty \quad \text{hence} \quad \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty.$$

2. $\frac{1}{x} \rightarrow -\infty$ as $x \rightarrow 0^-$ and hence $t_0 = -\infty$. Now we compute:

$$\lim_{t \rightarrow -\infty} e^t = 0 \quad \text{hence} \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0.$$

Example (3/3)



Change of variable in the limit:

- ▶ If $g(x) \rightarrow t_0$ as $x \rightarrow x_0$, $f(t) \rightarrow \ell$ as $t \rightarrow t_0$ and $g(x) \neq t_0$ eventually, then $f(g(x)) \rightarrow \ell$
- ▶ The hypothesis $g(x) \neq t_0$ is not needed if f is continuous at t_0 or if $t_0 = \pm\infty$
- ▶ Examples: $\log\left(\frac{2x^3+4x+1}{5(x+1)^3}\right) \rightarrow \log\frac{2}{5}$ as $x \rightarrow +\infty$;
 $e^{1/x} \rightarrow +\infty$ as $x \rightarrow 0^+$ and $e^{1/x} \rightarrow 0$ as $x \rightarrow 0^-$

Summary (1/2)

General strategy:

- ▶ The theorems on limits of functions follow from the theorems on sequences through the sequential definition of limit

Algebra of limits and sign preservation:

- ▶ Finite limits of sums, products and quotients
- ▶ The sign of the limit is transferred to the function (and vice versa), also for continuous functions

Summary (2/2)

Comparison and arithmetization of ∞ :

- ▶ Comparison theorem and its corollaries (infinitesimal times bounded is infinitesimal)
- ▶ Rules for $\ell \pm \infty$, $\ell \cdot \infty$, $\frac{\ell}{0}$, $\frac{\ell}{\infty}$, with the rule of signs

Change of variable:

- ▶ $\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t)$ with $t_0 = \lim_{x \rightarrow x_0} g(x)$